University of South Wales



Predicting Breast Cancer Detection using Machine Learning Algorithm

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Submitted by :
Name : Pritom Saha
Student ID: 30118440

Submitted to :
Module Tutor: Carl Jones

Abstract

Breast cancer is one of the diseases that causes a large number of deaths each year. It is the most common type of cancer and the leading cause of mortality among women around the world. This study focuses on the classification of medical images as either malignant or benign using machine learning techniques. Several machine learning algorithms, including Logistic Regression, Random Forest, Support Vector Machines, and Artificial Neural Network were implemented and evaluated. The data was meticulously prepared, including cleaning, feature selection, and splitting into training and testing sets. The performance of the models was assessed using metrics such as accuracy, precision, recall, and F1-score. The Random Forest classifier demonstrated superior performance in classifying the diagnosis and provides the highest accuracy (98.00%) with a lower error rate. The study concludes with a discussion of the efficacy of the implemented models, the advantages and disadvantages of the chosen algorithms, and potential improvements for future work. All work is done in the Anaconda environment, which uses the Python programming language, Scikit-learn, Keras, and TensorFlow library.

Contents

1 Introduction **1**

2 Proposed Research Model **1**

3 Methodology **1**

3.1 Dataset Overview: . . . 2

3.1.1 Data Cleaning and Preprocessing: . . . 3

3.2 Feature Engineering: . . . 3

3.3 Training Dataset: . . . 5

3.4 Applying Machine Learning Models: . . . 5

4 Results Analysis **5**

4.1 Analyzing All Algorithms . . . 6

4.1.1 Logistic Regression . . . 6

4.1.2 Decision Tree . . . 6

4.1.3 K-Nearest Neighbor(KNN) . . . 7

4.1.4 Random Forest . . . 8

4.1.5 Artificial Neural Network Model . . . 9

5 Discussion: **10**

6 Conclusions and Future Work **12**

References **13**

List of Figures

1 Proposed entire methodology . . . 1
2 Check All columns . . . 2
3 Drop ID . . . 2
4 Class Distribution of the Diagnosis Variable . . . 2
5 Statistical Report . . . 3
6 Null Values Check . . . 3
7 Check Duplicate Values . . . 3
8 Class imbalance using SMOTE . . . 3
9 Correlations between the features . . . 4
10 Feature scaling . . . 5
11 Train and Test Data Split . . . 5
12 Classification Report(a) and Confugion Matrix(b) of Logistic Regression . . . 6
13 Classification Report(a) and Confugion Matrix(b) of Decision Tree . . . 7
14 Classification Report(a) and Confugion Matrix(b) of KNN . . . 8
15 Classification Report(a) and Confugion Matrix(b) of Random Forest . . . 8
16 Classification Report(a) and Confugion Matrix(b) of ANN Model . . . 9
17 Model Performance Metrics Precision, Recall, F1-Score . . . 10
18 Receiver Operating Characteristic (ROC) Curve . . . 11

1 Introduction

Breast cancer is one of the most common cancers worldwide. Early detection through predictive analytics can improve treatment outcomes and survival rates. According to the World Health Organization (WHO), breast cancer accounts for approximately 12 % of all new cancer cases annually and remains a leading cause of cancer-related deaths among women. Every year, around 56,000 new instances of breast cancer are diagnosed in women, which equates to more than 150 cases every day. Breast cancer is the fourth most prevalent cause of cancer death and the second most common cause of cancer death among womenUK (2024). Breast cancer is the second leading cause of mortality among women in the United Kingdom, accounting for 15% of all female cancer deaths. It is not one of the top 20 causes of cancer death in men in the United Kingdom. In the United Kingdom, 99% of breast cancer deaths are female, with only 1% being male (2017-2019) UK (2025).

Machine learning (ML) offers a transformative approach to enhancing the accuracy, efficiency, and reliability of breast cancer detection. Using advanced algorithms and computational techniques, ML systems can analyze large amounts of data, identify patterns, and make predictions with a level of precision that exceeds conventional methods. These models can process diverse types of medical data, such as mammogram images, histopathological slides, and clinical metrics, to aid in the diagnosis of malignant (cancerous) or benign (non-cancerous) conditions. Using deep learning techniques on the Wisconsin Breast Cancer Database resulted in a 99.67% accurate diagnosis of breast cancer Khuriwal & Mishra (2018). The study compares five nonlinear machine learning methods, including Multilayer Perceptron (MLP), K-Nearest Neighbours (KNN), Classification and Regression Trees (CART), Gaussian Nave Bayes (NB), and Support Vector Machines (SVM), on Wisconsin Breast Cancer Diagnostic. MLP achieved a higher accuracy of 96.70% on training data compared to other four algorithms Al Bataineh (2019). This study compares the performance of various machine learning algorithms (SVM, Decision Tree, Naive Bayes, and k-NN) on the Wisconsin Breast Cancer datasets. SVM has the best accuracy (97.13%) and lowest error rate, according to experimental findings Asri et al. (2016).

2 Proposed Research Model

This section presents a comprehensive summary of our entire workflow, shown in Figure 1

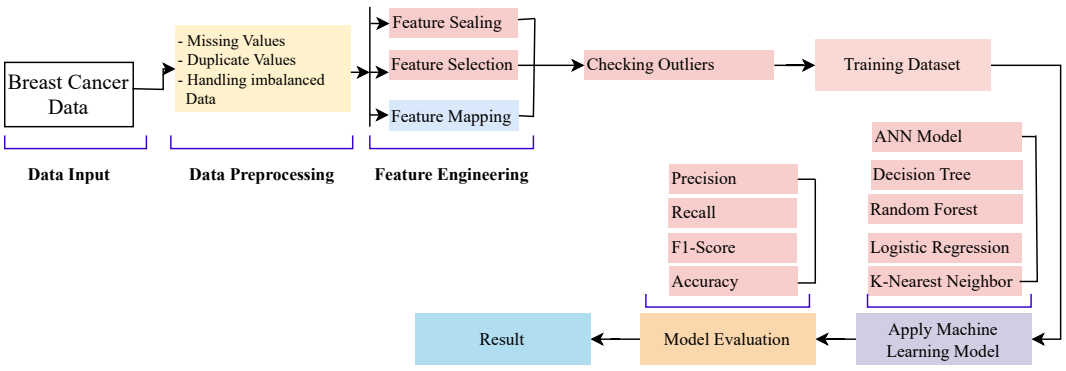


Figure 1: Proposed entire methodology

3 Methodology

In this section, we discuss Data Cleaning and Preprocessing, Feature Engineering, and Model Architecture.

3.1 Dataset Overview:

The pipeline begins with the input of breast cancer data. This data set could be sourced from publicly available data (Wisconsin Breast Cancer Dataset)Wolberg et al. (1993). Full code available on GitHub link Saha (2025)

Figure 2 Show all columns in our dataset using $b_c_dataset.columns$ methods.

```
b_c_dataset.columns
✓ 0.0s
Index(['id', 'diagnosis', 'radius_mean', 'texture_mean', 'perimeter_mean',
       'area_mean', 'smoothness_mean', 'compactness_mean', 'concavity_mean',
       'concave_points_mean', 'symmetry_mean', 'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se', 'area_se', 'smoothness_se',
       'compactness_se', 'concavity_se', 'concave_points_se', 'symmetry_se',
       'fractal_dimension_se', 'radius_worst', 'texture_worst',
       'perimeter_worst', 'area_worst', 'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave_points_worst',
       'symmetry_worst', 'fractal_dimension_worst'],
      dtype='object')
```

Figure 2: Check All columns

ID column is not needed for our analysis for this reason we drop the ID column from our dataset shown in Figure 3.

```
b_c_dataset = b_c_dataset.drop('id', axis=1)
✓ 0.0s
```

Figure 3: Drop ID

Data Distribution: The pie chart shows (Figure 4 (A)) the percentage distribution of these two categories, providing a quick overview of the amounts present in the dataset. whereas the bar chart shows (Figure 4 (B)) the overall number of benign and malignant tumors, including the exact number of instances in each category.

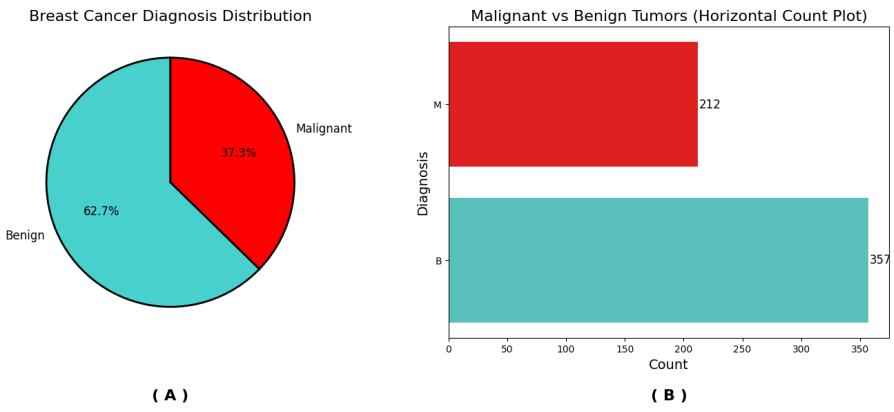


Figure 4: Class Distribution of the Diagnosis Variable

Describing Statistical Report : For checking statistical reports using $b_c_dataset.describe()$ method shown in figure 5. using this method we can easily check mean, min, max, count, std, and many other statistical variables.

```
b_c_dataset.describe()
```
✓ 0.0s Python

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave_points_mean	symmetry_mean	fractal_dimension_mean	...	radius_
count	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	...	569.0
mean	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	0.181162	0.062798	...	16.2
std	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	0.027414	0.007060	...	4.8
min	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	0.106000	0.049960	...	7.9
25%	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	0.161900	0.057700	...	13.0
50%	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	0.179200	0.061540	...	14.9
75%	15.780000	21.800000	104.100000	782.700000	0.105300	0.130400	0.130700	0.074000	0.195700	0.066120	...	18.7
max	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	0.304000	0.097440	...	36.0

8 rows × 30 columns

Figure 5: Statistical Report

3.1.1 Data Cleaning and Preprocessing:

Data cleaning and preprocessing ensure that the raw data is clean, consistent, and ready for training.

- **Check Null Values:** For checking null values using $b_c_dataset.isnull().sum()$ methods. After applying this method in the output shown in Figure 6 there are no null values or missing values in our dataset.

```
[12]: print(b_c_dataset.isnull().sum())

diagnosis                0
radius_mean              0
texture_mean             0
perimeter_mean           0
area_mean                0
smoothness_mean          0
compactness_mean         0
concavity_mean           0
concave_points_mean      0
symmetry_mean            0
fractal_dimension_mean   0
radius_se                0
texture_se               0
perimeter_se             0
area_se                  0
smoothness_se            0
compactness_se           0
concavity_se             0
concave_points_se        0
symmetry_se              0
fractal_dimension_se     0
radius_worst             0
texture_worst            0
perimeter_worst          0
area_worst               0
smoothness_worst         0
compactness_worst        0
concavity_worst          0
concave_points_worst     0
symmetry_worst           0
fractal_dimension_worst  0
dtype: int64
```

Figure 6: Null Values Check

- **Check Duplicate Values:** For checking Duplicate values in this project using $b_c_dataset.duplicated().sum()$ method. There are no duplicate values in our dataset shown in Figure 7.

```
b_c_dataset.duplicated().sum()
```
✓ 0.0s

```
0
```

Figure 7: Check Duplicate Values

- **Data Balanced:** Our dataset is not balanced look at Figure 4. The ratio of the target data is 62.7% Benign and 37.3% malignant. For handling imbalanced dataset here in this project using SMOTE methods shown in Figure 8. After apply this method our target class balanced.

```
smote = SMOTE(random_state=42)

# Apply SMOTE to the training data
X_resampled, y_resampled = smote.fit_resample(X, y)

# Check the class distribution after oversampling
print(y_resampled.value_counts())
```
✓ 0.3s

```
diagnosis
1    357
0    357
Name: count, dtype: int64
```

Figure 8: Class imbalance using SMOTE

3.2 Feature Engineering:

Feature engineering involves refining the data to enhance the model performance. Data normalization or standardization ensures that features are on a similar scale, preventing dominance by features with larger magnitudes. Techniques include feature scaling, correlation analysis, and checking outliers.

- **Correlation Matrix :** A correlation matrix is a valuable tool for visualizing and analyzing the linear relationships between variables in a dataset shown in figure 9. Red represents positive correlations and the deeper the shade, the stronger the positive correlation. White indicates no correlation (close to 0). Each cell in the matrix represents the correlation between two features, with values ranging from -1 to +1.

  i. +1 indicates Perfect positive correlation (as one variable increases, the other also increases proportionally).

  ii. -1 indicates Perfect negative correlation (as one variable increases, the other decreases proportionally).

  iii. 0 indicates No linear correlation (the variables do not have a linear relationship).

The diagonal values (correlation = 1.0) represent perfect correlation since each feature is perfectly correlated with itself. Strongly correlated features like radius_mean, perimeter_mean, and area_mean could be reduced to a representative subset, simplifying the model without significant loss of information.

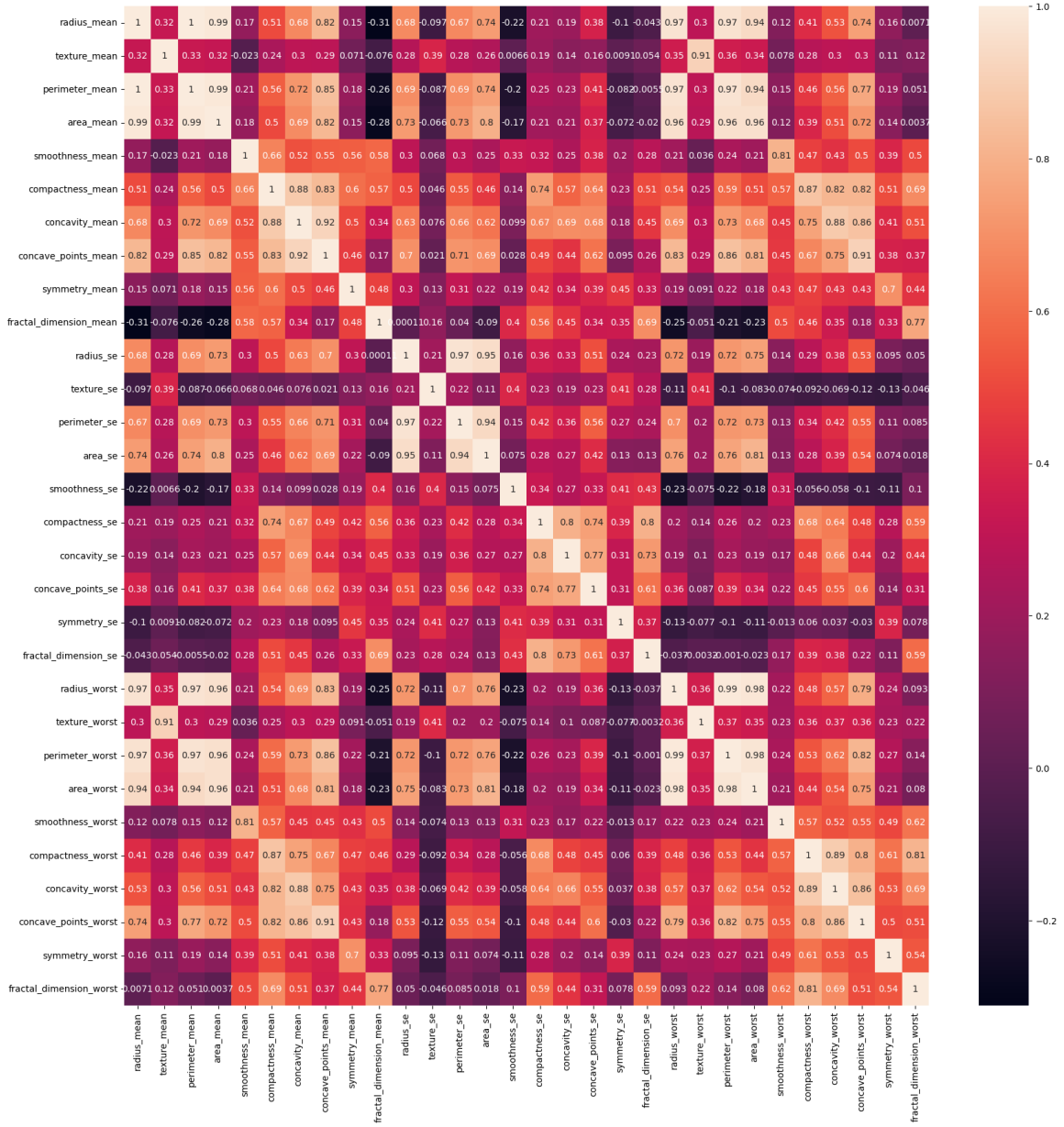


Figure 9: Correlations between the features

- **Feature scale :** Feature scaling is a preprocessing approach that standardizes or normalizes the dataset's independent features. For scaling here we use StandardScaler() methods shown in Figure 10

```
# Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_resampled)
```
✓ 0.0s

Figure 10: Feature scaling

3.3 Training Dataset:

Once preprocessing and feature engineering are completed, the clean and refined data is split into training and testing datasets. We divide training and testing sets in the ratio of 80:20 where 80 is the percentage of the training set, and 20 is the testing set shown in Figure 11. The training dataset is used to build machine learning models, while the testing dataset evaluates model performance.

```
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_resampled, test_size=0.2, stratify=y_resampled, random_state=42)
```
✓ 0.0s

Figure 11: Train and Test Data Split

3.4 Applying Machine Learning Models:

For this project we are using a total of 5 Models. We used Logistic Regression, Decision Tree, K-Nearest Neighbor, Random Forest and ANN model for predicting our data. We discussed all models in results Analysis section.

- **Model Evaluation:** The performance of the machine learning models is assessed using the following metrics:

  i. Precision: Measures the proportion of correctly identified positive cases.

  ii. Recall: Measures the proportion of actual positive cases that were correctly identified.

  iii. F1-Score: The harmonic mean of precision and recall, balancing both metrics.

  iv. Accuracy: The percentage of correctly classified instances out of the total instances.

4 Results Analysis

The final result includes the evaluation metrics and identifies the best-performing model based on its ability to predict breast cancer accurately. Insights from the evaluation are used to determine the model's reliability and potential for deployment in clinical settings.

4.1 Analyzing All Algorithms

4.1.1 Logistic Regression

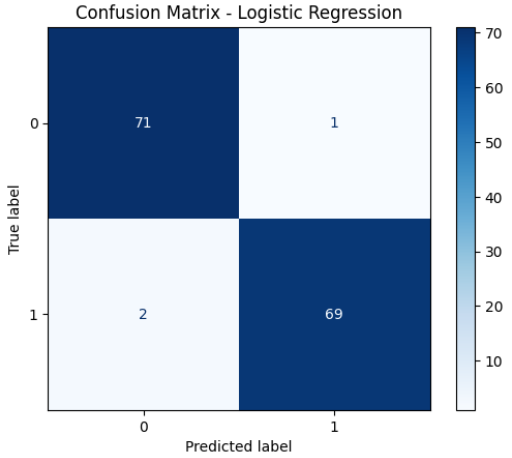
The Logistic Regression model is a widely used classification algorithm based on statistical principles. It is also applicable to multiclass classification tasks with some modifications Uses a sigmoid function to map predicted values to a probability range between 0 and 1. Logistic Regression excels at binary classification tasks like distinguishing between benign and malignant tumors. It provides probabilities for each class, helping interpret the confidence of predictions. The model is computationally efficient and works well when relationships between features and outcomes are approximately linear. Coefficients from the model can help identify which features are most relevant to classification. Classification report 12(a) and confusion matrix 12(b) show model performance.

1. Accuracy: The model achieves 98%, indicating reliable overall performance.
2. Precision: For class 0 (Benign), the model predicts a tumor as benign (Class 0), 97% of the time it is correct. And for class 1 (Malignant) the model predicts a tumor as malignant (Class 1), 98% of the time it is accurate.
3. Recall (Sensitivity): For class 0 (Benign) 99%, meaning most benign tumors are correctly identified.And class 1 (Malignant) 97%, indicating room for improvement in identifying malignant tumors to minimize false negatives.
4. F1-Score: The F1-scores 98% confirm a good balance between precision and recall.

Classification Report - Logistic Regression

	precision	recall	f1-score	support
0	0.97	0.99	0.98	72
1	0.99	0.97	0.98	71
accuracy			0.98	143
macro avg	0.98	0.98	0.98	143
weighted avg	0.98	0.98	0.98	143

(a)



(b)

Figure 12: Classification Report(a) and Confugion Matrix(b) of Logistic Regression

4.1.2 Decision Tree

A Decision tree is a supervised learning algorithm primarily used for classification and regression tasks. It works by splitting the dataset into smaller subsets based on feature values, creating a tree-like structure where each branch represents a decision rule, and each leaf node corresponds to an outcome. In breast cancer prediction, a decision tree would split the dataset using features like tumor size, texture, and shape to classify whether a tumor is Benign (B) or Malignant (M). Easy to visualize and understand the decision-making process. Classification report 13b(a) and confusion matrix 13b(b) show model performance.

i. Precision: Proportion of true positive predictions to the total predicted positives. For class 0: 0.97 (Benign cases are correctly predicted with 97% precision.) For class 1: 0.96 (Malignant cases are correctly predicted with 96% precision.)

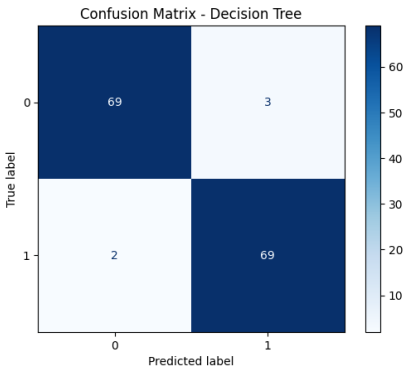
ii. Recall (Sensitivity): Proportion of true positives to the total actual positives. For class 0: 0.96 (96 % of benign cases are correctly identified.) For class 1: 0.97 (97% of malignant cases are correctly identified.)

iii. F1-Score: Harmonic mean of precision and recall, balancing their trade-off. Both classes achieve an F1-score of 0.97, indicating strong performance.

Classification Report - Decision Tree

	precision	recall	f1-score	support
0	0.97	0.96	0.97	72
1	0.96	0.97	0.97	71
accuracy			0.97	143
macro avg	0.97	0.97	0.97	143
weighted avg	0.97	0.97	0.97	143

(a)



(b)

Figure 13: Classification Report(a) and Confugion Matrix(b) of Decision Tree

4.1.3 K-Nearest Neighbor(KNN)

K-Nearest Neighbors (KNN) is a simple and effective machine learning algorithm primarily used for classification and regression tasks. The KNN model achieves high precision, recall, and F1-scores for both classes, indicating balanced performance.Misclassifications are minimal, with 4 samples of Class 0 predicted as Class 1 and 2 samples of Class 1 predicted as Class 0. Classification report 14(a) and confusion matrix 14(b) show model performance.

i. Precision: Measures the proportion of true positives among the predicted positives. For class 0: 0.97 (Benign cases are correctly predicted with 97% precision.) For class 1: 0.95 (Malignant cases are correctly predicted with 95% precision.)

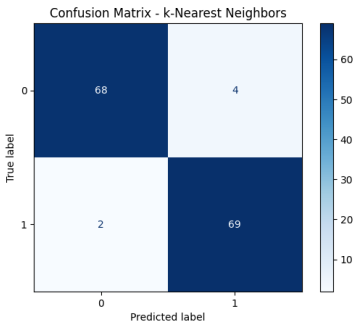
ii. Recall: Measures the proportion of true positives among the actual positives. For class 0: 0.94 (94 % of benign cases are correctly identified.) For class 1: 0.97 (97% of malignant cases are correctly identified.)

iii. F1-Score: The harmonic mean of precision and recall, balancing both metrics. Both classes achieve an F1-score of 0.96, indicating strong performance.

Classification Report - k-Nearest Neighbors

	precision	recall	f1-score	support
0	0.97	0.94	0.96	72
1	0.95	0.97	0.96	71
accuracy			0.96	143
macro avg	0.96	0.96	0.96	143
weighted avg	0.96	0.96	0.96	143

(a)



(b)

Figure 14: Classification Report(a) and Confugion Matrix(b) of KNN

4.1.4 Random Forest

The Random Forest is a robust tree-based ensemble learning method. It combines multiple decision trees, aggregating their outputs (via majority voting for classification) to enhance accuracy, generalization, and reduce overfitting. Random Forest mitigates overfitting by combining multiple decision trees, resulting in more reliable predictions. It identifies the most significant features that contribute to predictions, useful for medical datasets like breast cancer detection. Random Forest performs well even with slight class imbalance by focusing on individual class performance. The Random Forest model is a highly accurate and interpretable algorithm for breast cancer detection. Classification report 15b(a) and confusion matrix 15b(b) show model performance.

i. Precision: For class 0: 0.96 (Benign cases are predicted with 96% precision.) and class 1: 1.00 (Malignant cases are perfectly predicted without any false positives.)

ii. Recall (Sensitivity): For class 0: 1.00 (100% of benign cases are correctly identified.) and class 1: 0.96 (96% of malignant cases are correctly identified.)

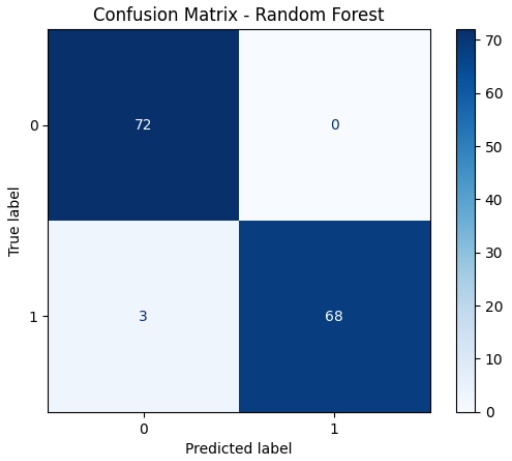
iii. F1-Score: For class 0: 0.98 (Balanced performance for benign cases.) and class 1: 0.98 (Balanced performance for malignant cases.)

Overall Accuracy: 98% of the total predictions are correct.

Classification Report - Random Forest

	precision	recall	f1-score	support
0	0.96	1.00	0.98	72
1	1.00	0.96	0.98	71
accuracy			0.98	143
macro avg	0.98	0.98	0.98	143
weighted avg	0.98	0.98	0.98	143

(a)



(b)

Figure 15: Classification Report(a) and Confugion Matrix(b) of Random Forest

4.1.5 Artificial Neural Network Model

An Artificial Neural Network (ANN) is a computational model inspired by the human brain. It is capable of learning complex patterns in data, making it an excellent choice for breast cancer detection where subtle differences in features can signify different diagnoses.ANN handles nonlinear relationships between features and class labels, which are common in medical data. With sufficient data and preprocessing, ANNs outperform traditional models by capturing intricate patterns. It contains the classification performance metrics and the confusion matrix for an Artificial Neural Network (ANN) model. Classification report 16b(a) and confusion matrix 16b(b) show model performance.

i. Precision: Measures the proportion of true positives out of all predicted positives for each class.For class 0: 0.96 (Benign cases predicted with high precision.) and class 1: 0.99 (Malignant cases predicted with very high precision.)

ii. Recall: Measures the proportion of true positives out of all actual positives for each class.For class 0: 0.99 (99% of benign cases are correctly identified.) and class 1: 0.96 (96% of malignant cases are correctly identified.)

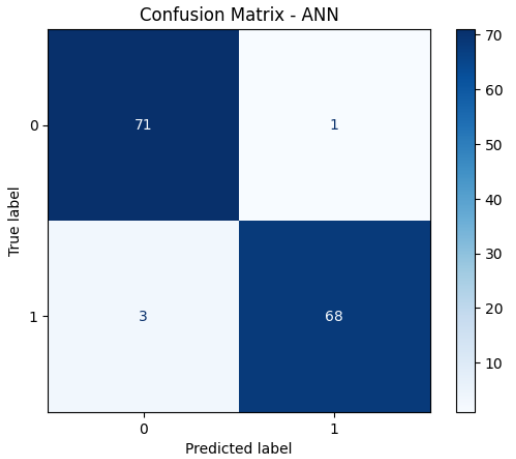
iii. F1-Score: The harmonic mean of precision and recall, representing a balance between the two.Both classes achieve a balanced F1-score of 0.97.

Overall Accuracy: 97% of the total predictions are correct.

Classification Report - ANN

	precision	recall	f1-score	support
0	0.96	0.99	0.97	72
1	0.99	0.96	0.97	71
accuracy			0.97	143
macro avg	0.97	0.97	0.97	143
weighted avg	0.97	0.97	0.97	143

(a)



(b)

Figure 16: Classification Report(a) and Confugion Matrix(b) of ANN Model

5 Discussion:

Figure 17 displays a bar chart comparing the precision, recall, and F1-score of different machine learning models.

- Precision: Measures the proportion of correctly predicted positive observations to the total predicted positives. High precision across all models indicates reliable predictions of the positive class.
- Recall: Measures the proportion of correctly predicted positive observations to all actual positives. Similar values to precision suggest that the models also capture most of the positive cases.
- F1-Score: Combines precision and recall into a single metric. The values reflect the balance between precision and recall.

Random Forest outperforms the other models in all three metrics, making it the most effective model for this specific classification task. While Logistic Regression and KNN perform adequately, they lag behind tree-based methods and ANN. The metrics for all models are closely aligned, showing that they perform well overall.

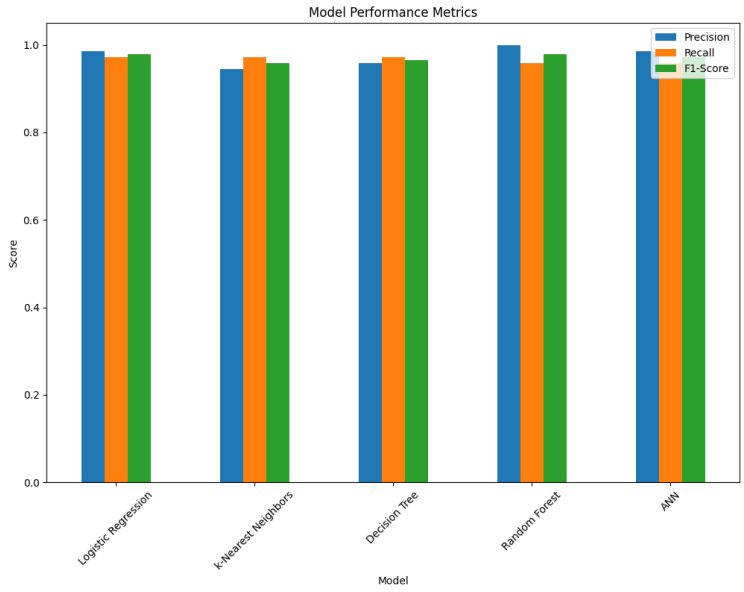


Figure 17: Model Performance Metrics Precision, Recall, F1-Score

- Logistic Regression : Precision, recall, and F1-score are slightly lower than the other models.
- K-Nearest Neighbors (KNN): Performs comparably to Logistic Regression but shows a slight improvement in all three metrics.
- Decision Tree: Marginally better performance than Logistic Regression and KNN.
- Random Forest: Precision, recall, and F1-score are higher, indicating better generalization and robustness compared to Decision Tree.
- Artificial Neural Network (ANN): Consistently achieves the highest precision, recall, and F1-score among the models, demonstrating superior performance.

The Receiver Operating Characteristic (ROC)curve evaluates the trade-off between the true positive rate (TPR) and the false positive rate (FPR) for each model across various classification thresholds.

Logistic Regression and Random Forest: Both achieve perfect scores (AUC = 1.00), indicating no trade-offs between sensitivity and specificity.

KNN, Decision Tree, and ANN: Perform slightly worse than Logistic Regression and Random Forest but still achieve AUC = 0.99, signifying strong performance. Curves Proximity to Top-Left Corner: All curves are tightly clustered near the top-left, confirming the models perform well with minimal false positives.

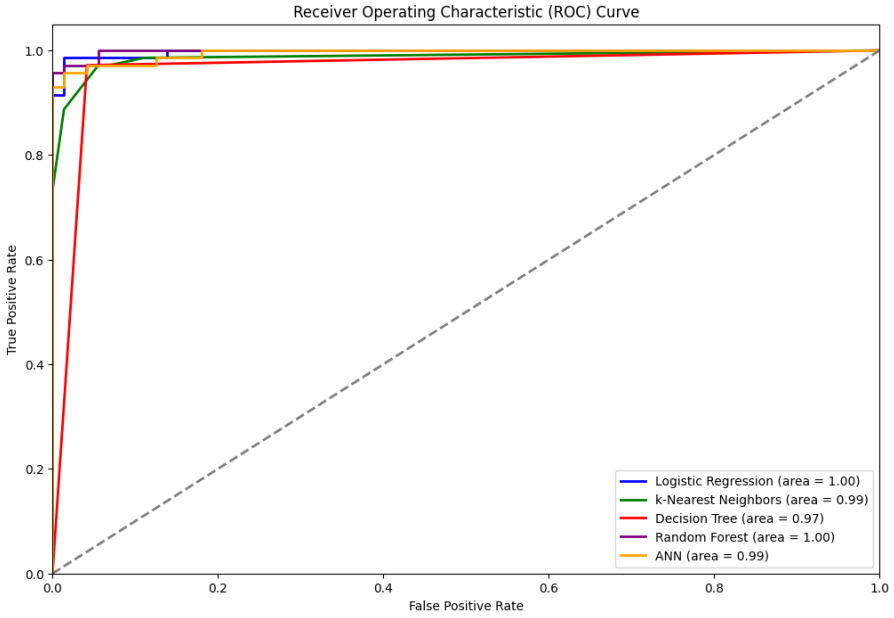


Figure 18: Receiver Operating Characteristic (ROC) Curve

6 Conclusions and Future Work

In this report, we used four machine learning and deep learning methods artificial neural network for detecting breast cancer dataset. After the implementation of all methods, we have achieved 98.00% accuracy on both logistic regression and random forest models. Compare and assess findings based on confusion matrix, accuracy, sensitivity, precision, and AUC to determine the most precise, dependable, and accurate machine learning algorithms. The algorithms were developed in Python using the scikit-learn module in the Anaconda environment. Our analysis revealed that the Random Forest algorithm outperformed all other methods, achieving 95.77% recall, 100.00% precision, 97.84% F1-score and 100.00% AUC. In this report, we used all features for the diagnosis of breast cancer. In the future, will apply advanced feature selection methods to reduce features. we will apply Convolution Neural Network(CNN) and try to achieve the best accuracy.

Future Work Ensure that the data is of good quality and enough for training. Examining strategies including oversampling, undersampling, and utilising multiple assessment metrics to account forthe class distribution and combining predictions from many models to improve overall accuracy and robustness. Techniques such as k-fold cross-validation were used to obtain a more reliable estimate of the model's performance. Using tactics to prevent overfitting, such as regularisation techniques, early halting during training, or ensemble approaches that naturally reduce overfitting, such as Random Forests. In addition, we will use hyperparameter optimization to optimize our outcomes.

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